

OPTIMIZING TELEMARKETING TERM DEPOSIT CONVERSION USING PREDICTIVE ANALYTICS



Project Team:

- Karan Paul Arora (Lead): Roll no - iimsi_edge_2511072
- Ruchi Kumari: Roll no - iimsi_edge_2511100
- Aryan Amit Singh: Roll no - iims_edge_2511006
- Parth Yogesh Sawant: Roll no - iims_edge_2511188
- Deshna Chopra: Roll no - iimsi_edge_2511193
- Meeta Rani: Roll no - iimsi_edge-2511070

Submitted in partial fulfillment of the requirements for the completion of EDGE – Essentials of Data, GenAI & Execution

Certificate

This is to certify that the project entitled **"Optimizing Telemarketing Term Deposit Conversion Using Predictive Analytics"** has been carried out by the above-mentioned team members as part of the requirements of the **EDGE – Essentials of Data, GenAI & Execution** programme.

The work presented in this report is original and has been completed under appropriate academic guidance. The project demonstrates the practical application of predictive analytics and logistic regression techniques for improving telemarketing efficiency in banking by identifying customers with a higher probability of subscribing to term deposits.



Declaration

We hereby declare that the project entitled **"Optimizing Telemarketing Term Deposit Conversion Using Predictive Analytics"** is our original work.

The report has been prepared solely for academic purposes under the EDGE programme. All information collected from secondary sources has been properly acknowledged. The statistical analysis, interpretations, and recommendations presented in this report are based on the dataset used in this study.

We further declare that this work has not been submitted previously for the award of any other degree, diploma, or certification.

Acknowledgement

We express our sincere gratitude to our faculty members, mentors, and the EDGE programme coordinators for providing us with the opportunity to undertake this project.

We would also like to thank the creators of the Bank Marketing Dataset available on Kaggle for making the dataset publicly available for academic research.

We are grateful to our teammates for their continuous cooperation, valuable discussions, and dedicated efforts throughout the project. Their contribution played a significant role in the successful completion of this study.

Finally, we thank everyone who directly or indirectly supported us during the preparation of this project report.

Table of Contents

Chapter	Page No:
Certificate Declaration Acknowledgement Table of Content	2-5
Executive Summary	6
Business Problem & Operational Challenge and Strategic Objectives	7-8
Research Objectives	8
Project Theme	9
Dataset Description 1. Dataset Overview 2. Dataset Source 3. Dataset Characteristics 4. Variables Included 5. Target Variable	10-12
Methodology 1. Research Methodology & Framework 2. Data Setup & Cleaning 3. Feature Encoding 4. Data Partitioning 5. Sequential Stepwise Modelling	13-15
Correlation Analysis • Feature Engineering • Variable selection strategy	16-17
Model Evolution and Key Findings	18
Model Performance Metrics	19
Significant Predictors (Model 14)	20
Model Evaluation • Classification Accuracy	21
Performance Evaluation and Threshold Strategy	22
Confusion Matrix	23
Business Interpretation	24
Overall Findings	25
Business Recommendations	26-27
Conclusion	28
References & Appendix	29

Executive Summary

- ✓ Telemarketing remains one of the most widely used direct marketing strategies in the banking industry for promoting financial products such as term deposits.
- ✓ Conventional telemarketing campaigns often rely on indiscriminate mass calling, resulting in low conversion rates, excessive operational costs, inefficient utilization of call center resources, and increased customer dissatisfaction.
- ✓ To overcome these challenges, organizations are increasingly adopting predictive analytics techniques that enable data-driven customer targeting and improve campaign effectiveness.
- ✓ This project presents a predictive analytics framework designed to optimize telemarketing term deposit conversion using Binary Logistic Regression.
- ✓ The analysis is based on the Bank Marketing Dataset, which contains 41,188 customer records with demographic, financial, campaign-related, and macroeconomic variables.
- ✓ The primary objective of the study is to identify customers who are most likely to subscribe to a term deposit before initiating telemarketing contact.
- ✓ A sequential stepwise Logistic Regression approach was adopted to develop multiple predictive models.
- ✓ Fourteen models were evaluated using Akaike Information Criterion (AIC) and Residual Deviance (RD) to determine the most efficient combination of predictor variables.
- ✓ The final model demonstrated substantial improvement over the initial models while maintaining statistical significance and avoiding multicollinearity.
- ✓ The findings reveal that previous campaign outcomes, customer demographics, contact method, macroeconomic indicators, and prior customer interactions significantly influence subscription decisions.
- ✓ Performance evaluation using different probability thresholds showed that a classification threshold of 0.60 achieved an overall prediction accuracy of approximately 89.83%, offering an effective balance between identifying potential subscribers and minimizing unnecessary calls.
- ✓ The proposed predictive framework enables financial institutions to prioritize high-probability customers instead of relying on random telemarketing campaigns.
- ✓ By implementing this approach, banks can reduce operational costs, improve agent productivity, increase conversion rates, and enhance overall customer experience.
- ✓ The study demonstrates how predictive analytics can transform traditional telemarketing into a more efficient, data-driven decision-support system for customer acquisition.

Business Problem

- ✓ The banking industry extensively uses telemarketing campaigns to promote financial products such as term deposits.
- ✓ Although telemarketing enables direct communication with potential customers, traditional campaigns often rely on mass calling strategies, where every customer in the database is contacted without evaluating their likelihood of subscribing.
- ✓ This approach results in low conversion rates, inefficient use of organizational resources, increased operational costs, and customer dissatisfaction due to repeated and irrelevant calls.
- ✓ In a highly competitive banking environment, organizations require data-driven decision-making techniques that can accurately identify prospective customers before initiating contact.
- ✓ Predictive analytics provides an effective solution by utilizing historical customer information, previous campaign outcomes, demographic characteristics, financial attributes, and macroeconomic indicators to estimate the probability of customer subscription.
- ✓ The primary business challenge addressed in this study is improving the effectiveness of telemarketing campaigns by replacing random customer selection with predictive customer targeting.
- ✓ Rather than contacting every customer, the proposed predictive model enables banks to prioritize customers who are more likely to subscribe to a term deposit, thereby increasing marketing efficiency and reducing unnecessary operational expenditure.
- ✓ The implementation of predictive analytics in telemarketing campaigns offers several business advantages, including improved campaign performance, higher conversion rates, better allocation of call center resources, reduced customer fatigue, and enhanced customer relationship management.

Operational Challenge and Strategic Objectives

The core business problem identified is the reliance on "Mass Random Calling," which exhausts call center capacity and yields poor results. The strategic shift toward "Data-Driven Target Precision" addresses four primary operational hurdles:

- ✓ **Low Conversion Rates:** Sales agents currently spend significant time on clients with zero intention of opening term deposits.
- ✓ **Resource Waste:** Indiscriminate cold-calling of entire databases leads to massive operational overhead.
- ✓ **Customer Fatigue:** Repeated contact of uninterested clients results in brand irritation and customer drop-off.
- ✓ **Strategic Goal:** The project aims to use historical campaign and economic data to score customers, ensuring agents only contact high-probability leads to lower acquisition costs.

Research Objectives

The study aims to develop a predictive analytics framework capable of improving telemarketing efficiency by accurately identifying customers who are more likely to subscribe to a term deposit.

The specific objectives are:

- To develop a Binary Logistic Regression model for predicting customer subscription to term deposits.
- To identify the demographic, financial, campaign-related, and macroeconomic variables that significantly influence customer subscription decisions.
- To compare multiple sequential logistic regression models using Akaike Information Criterion (AIC) and Residual Deviance (RD).
- To evaluate model performance using classification accuracy and probability threshold analysis.
- To recommend a predictive customer targeting strategy that improves telemarketing efficiency while reducing operational costs.
- To provide practical business recommendations for implementing predictive analytics in banking telemarketing campaigns.

Project Theme

- ✓ The project focuses on applying predictive analytics to optimize banking telemarketing operations through intelligent customer targeting.
- ✓ Instead of relying on conventional mass calling strategies, the proposed framework utilizes historical customer data and logistic regression modelling to estimate the probability of customer subscription before initiating contact.
- ✓ The study combines statistical modelling with business decision-making to demonstrate how machine learning techniques can improve marketing efficiency.
- ✓ By identifying customers with a higher likelihood of subscription, financial institutions can prioritize valuable leads, improve campaign effectiveness, reduce unnecessary customer contacts, and maximize return on marketing investment.
- ✓ The project highlights the practical application of predictive analytics as a decision-support system that enables organizations to transition from traditional marketing approaches toward data-driven customer acquisition strategies.

Dataset Description

Dataset Overview

- ✓ The study utilizes the **Bank Marketing Dataset**, a widely recognized dataset used for predictive modelling and customer behaviour analysis in the banking sector.
- ✓ The dataset contains **41,188 customer records**, where each observation represents an individual customer who was contacted during a direct telemarketing campaign promoting term deposit subscriptions.
- ✓ The primary objective of the analysis is to predict whether a customer will subscribe to a term deposit before initiating telemarketing contact.
- ✓ The target variable used in this study is: **y**

- **Yes (1)** – Customer subscribed to a term deposit.
- **No (0)** – Customer did not subscribe.

The dataset contains customer demographic information, financial status, previous marketing campaign details, communication information, and macroeconomic indicators that influence customer decisions.

Dataset Source

The dataset used in this study was obtained from the publicly available **Bank Marketing Dataset**.

Source: UCI Machine Learning Repository

Dataset Name: Bank Marketing Dataset (bank-additional-full.csv)

Alternative Source: Kaggle

The dataset has been extensively used in predictive analytics, machine learning, and banking research for customer subscription prediction.

Dataset Characteristics

Dataset Information

Property	Description
Industry	Banking
Dataset name	Bank Marketing Dataset
Number of Records	41,188
Target Variable	Customer Subscription (y)
Target Classes	Yes / No
Missing Values	None
Problem Type	Binary Classifications
Technique used	Binary Logistic Regressions

Variables Included

The dataset contains variables belonging to four major categories.

A. Demographic Variables

- Age
- Job
- Marital Status
- Education

These variables describe the customer's personal background and socioeconomic characteristics.

B. Financial Variables

- Housing Loan
- Personal Loan
- Credit Default

These variables indicate the customer's financial obligations and borrowing profile.

C. Campaign Variables

- Contact Type
- Campaign Count
- Previous Campaign Outcome
- Number of Days Since Last Contact (pdays)

These variables describe the customer's interaction history with previous telemarketing campaigns.

D. Economic Indicators

- Consumer Price Index (CPI)
- Consumer Confidence Index
- Euro Interbank offered rate
- Employment Variation Rate

  These macroeconomic indicators capture the economic environment during the campaign period and influence customer investment decisions.

Target Variable

Variable	Description
y	**Customer subscribed to term deposit (Yes/No)**

The target variable forms the basis of the Binary Logistic Regression model developed in this study.

Chapter Summary

The Bank Marketing Dataset provides comprehensive information on customer demographics, financial characteristics, previous marketing campaigns, and macroeconomic conditions. Its balanced combination of customer-specific and external economic variables makes it highly suitable for developing predictive models that support data-driven telemarketing decisions.

Methodology

4.1 Research Methodology

This study follows a structured predictive analytics framework to improve the effectiveness of telemarketing campaigns for term deposit subscriptions. The methodology combines data preprocessing, exploratory data analysis, feature engineering, and Binary Logistic Regression modelling to identify customers who are most likely to subscribe to a term deposit before telemarketing contact.

The overall analytical process consists of the following stages:

1. Data Collection
2. Data Cleaning and Preprocessing
3. Feature Encoding
4. Exploratory Data Analysis (EDA)
5. Train-Test Data Partitioning
6. Logistic Regression Model Development
7. Model Selection using AIC and Residual Deviance
8. Model Evaluation
9. Business Interpretation and Recommendations

4.1 Research Methodology Framework

Dataset Collection

↓

Data Cleaning & Preparation

↓

Feature Encoding

↓

Exploratory Data Analysis

↓

Train-Test Split (70:30)

↓

Sequential Logistic Regression (Model 1 → Model 14)

↓

Model Evaluation (AIC, RD, Accuracy, VIF)

↓

Business Interpretation

↓

Recommendations

The project followed a rigorous five-step analytical process to ensure model validity and predictive power.

1. Data Setup & Cleaning

The dataset required minimal cleaning because no missing values were present.

One important preprocessing step involved removing the variable:

1. Duration - The duration variable represents the duration of the phone call.

Although this variable has a strong relationship with customer subscription, it was intentionally removed because it becomes available only after the telemarketing call has already been completed.
Including this variable would introduce data leakage, making the predictive model unrealistic for practical implementation.
Therefore, duration was excluded before model development.

2. Target Variable Preparation

The dependent variable - **y**
contains two categories:

- Yes
- No

For Logistic Regression, these values were converted into binary form.

Original Value	Encoded Value
Yes	1
No	0

This transformation enabled Binary Logistic Regression to estimate the probability of customer subscription.

3. Feature Encoding

The dataset contains several categorical variables that cannot be processed directly by Logistic Regression.

Therefore, appropriate encoding techniques were applied.

Binary Variables

Variables such as

- Housing Loan
- Personal Loan
- Default

were converted into

- 1 = Yes
- 0 = No

using Label Encoding.

4. Dummy Variable Encoding

Multi-category variables including

- Job
- Education
- Marital Status
- Contact
- Month
- Previous Outcome

were transformed using **Dummy Variable Encoding** through the **fastDummies** package in R.

Each category was converted into a separate binary variable while excluding one reference category to avoid the Dummy Variable Trap.

5. Data Partitioning

To evaluate predictive performance on unseen observations, the dataset was divided into training and testing subsets.

Dataset	Percentage	Observations
Training Dataset	70%	28,832
Testing Dataset	30%	12,356

The training dataset was used for model development, while the testing dataset was used to evaluate prediction accuracy.

Class Distribution

The target variable showed the following distribution.

Customer Response	Frequency
No	36,548
Yes	4,640

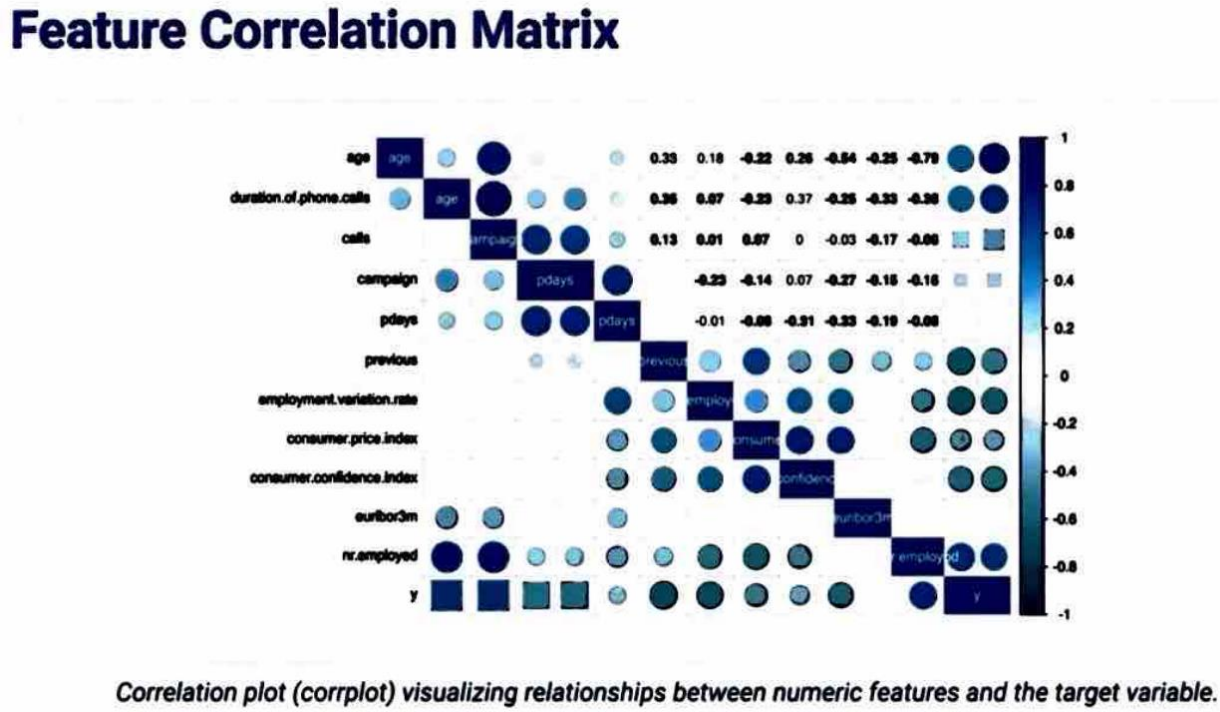
The dataset is imbalanced, with most customers not subscribing to the term deposit.

This imbalance was considered during model interpretation.

Correlation Analysis

Correlation analysis was performed among numerical variables to identify relationships and detect possible multicollinearity.

The analysis indicated that most predictor variables had acceptable correlation levels, supporting the development of Logistic Regression models.



Correlation plot (corrplot) visualizing relationships between numeric features and the target variable.

4.4 Feature Engineering

Feature engineering improves predictive performance by selecting and transforming relevant variables before model development.

In this study, feature engineering primarily involved selecting meaningful customer characteristics and campaign variables that influence subscription behaviour.

The selected variables included:

Demographic Variables a) Age b) Job c) Marital Status d) Education	Financial Variables a) Housing Loan b) Personal Loan c) Default
Campaign Variables a) Contact Method b) Campaign Count c) Previous Outcome d) Pdays	Economic Variables a) Consumer Price Index b) Consumer Confidence Index c) Employment Variation Rate d) Euro Interbank offered rate

Variable Selection Strategy

Instead of including every variable simultaneously, a **Sequential Stepwise Logistic Regression** approach was adopted.

Variables were added progressively across multiple models to evaluate their contribution toward improving predictive performance.

The quality of each model was assessed using:

- Akaike Information Criterion (AIC) - A metric used to compare model quality, rewarding goodness of fit while penalizing over-complexity.
- Residual Deviance (RD) - A measure of how well the model fits the data.
- Statistical Significance (p-value)
- Variance Inflation Factor (VIF) - A Variance Inflation Factor (VIF) of < 5 was maintained to ensure independent variables were not overly correlated.

This systematic process ensured that only statistically significant variables were retained in the final model.

Model Evolution and Key Findings & Analysis

Model: Binary Logistic regression

Objective:

The objective of this study is to develop a predictive model capable of identifying customers who are most likely to subscribe to a term deposit before telemarketing contact.

By estimating the probability of customer subscription, the model enables banks to optimize telemarketing campaigns, improve conversion rates, and reduce unnecessary operational costs.

Unlike traditional mass-calling strategies, the proposed model supports a data-driven approach to customer selection, allowing marketing teams to prioritize high-potential customers and improve campaign efficiency.

Dependent Variable

Target Variable (y)

The dependent variable represents whether a customer subscribed to a term deposit.

Category	Coding
No	0
Yes	1

Binary Logistic Regression was selected because the response variable consists of two mutually exclusive categories.

Independent Variables

The final predictive model considered customer demographic, financial, campaign-related, and macroeconomic variables.
The principal predictor variables include:

- Age
- Job
- Marital Status
- Education
- Housing Loan
- Personal Loan
- Contact Type
- Campaign
- Previous Outcome
- Pdays
- Consumer Price Index (CPI)
- Consumer Confidence Index
- Euro Interbank offered rate
- Employment Variation Rate

The modeling process showed a consistent decrease in both AIC and RD as more relevant variables were added, indicating improved predictive power.

Model Performance Metrics

Model	AIC	Residual Deviance (RD)
Model 1	17796	17792
Model 4	17002	16992
Model 7	16741	16725
Model 14 (Final)	16548	16518

Interpretation of Model Evolution

The comparison demonstrates a consistent reduction in both Akaike Information Criterion (AIC) and Residual Deviance (RD) as additional significant variables were incorporated into the regression model.

A lower AIC indicates a more efficient model by balancing goodness of fit with model complexity, while a lower Residual Deviance signifies an improved ability to explain the observed customer responses.

Among all fourteen models, **Model 14** achieved the lowest AIC and Residual Deviance, indicating the best overall predictive performance.

Multicollinearity Assessment

Prior to interpreting the regression coefficients, Variance Inflation Factor (VIF) analysis was conducted to examine multicollinearity among the predictor variables.

Decision Rule

- VIF < 5 → No serious multicollinearity
- VIF > 5 → Potential multicollinearity concern

The VIF values obtained for the final model were all below the acceptable threshold, indicating that the independent variables contributed unique information and did not exhibit problematic multicollinearity.

This confirms the statistical reliability of the regression coefficients.

Significant Predictors (Model 14)

The final complex model identified several variables with high statistical significance ($p < 2e\text{-}16$):

Economic Indicators: -	**Customer History: -**
The **Euro Interbank Offered Rate** emerged as one of the strongest predictors of customer subscription. Changes in interest rates significantly influenced customers' willingness to invest in term deposits, demonstrating the importance of macroeconomic conditions in marketing decisions. Similarly, the **Consumer Price Index (CPI)** significantly affected customer behaviour, indicating that inflationary conditions influence investment preferences.	Previous marketing interactions played a critical role in predicting customer decisions. Customers who had responded positively during previous campaigns exhibited a substantially higher probability of subscribing again. Likewise, the variable **pdays**, representing the number of days since the previous contact, significantly influenced subscription probability, suggesting that the timing of customer contact is an important determinant of campaign success.
Contact Method: -	**Demographic Variables: -**
The analysis revealed that customers contacted through **cellular communication** were significantly more likely to subscribe than those contacted via traditional telephone channels. This finding indicates that selecting the appropriate communication channel can substantially improve campaign performance.	Customer occupation also influenced subscription behaviour. Individuals belonging to the **retired** and **student** categories demonstrated comparatively higher probabilities of subscribing to term deposits. Education level also exhibited statistical significance, particularly among customers possessing university degrees.

Model Evaluation

The final Binary Logistic Regression model was evaluated using multiple statistical and classification performance measures to assess its predictive capability. The evaluation focused on determining how accurately the model distinguished customers who subscribed to a term deposit from those who did not.

The performance of the model was assessed using:

- Classification Accuracy
- Confusion Matrix
- Classification Threshold Analysis
- Sensitivity
- Specificity
- False Positives and False Negatives

These measures provide a comprehensive understanding of the model's effectiveness in supporting business decision-making.

Classification Accuracy

The final model was tested using the unseen testing dataset consisting of **12,356 customer records**.

The model achieved an overall prediction accuracy of approximately **89.96%** when a probability threshold of **0.60** was applied.

This level of accuracy demonstrates that the developed model effectively distinguishes potential subscribers from non-subscribers and is suitable for practical implementation in banking telemarketing campaigns.

Performance Evaluation and Threshold Strategy

Logistic Regression produces probabilities rather than direct classifications. Therefore, a probability threshold must be selected to determine whether a customer should be classified as a potential subscriber.

Two thresholds were evaluated:

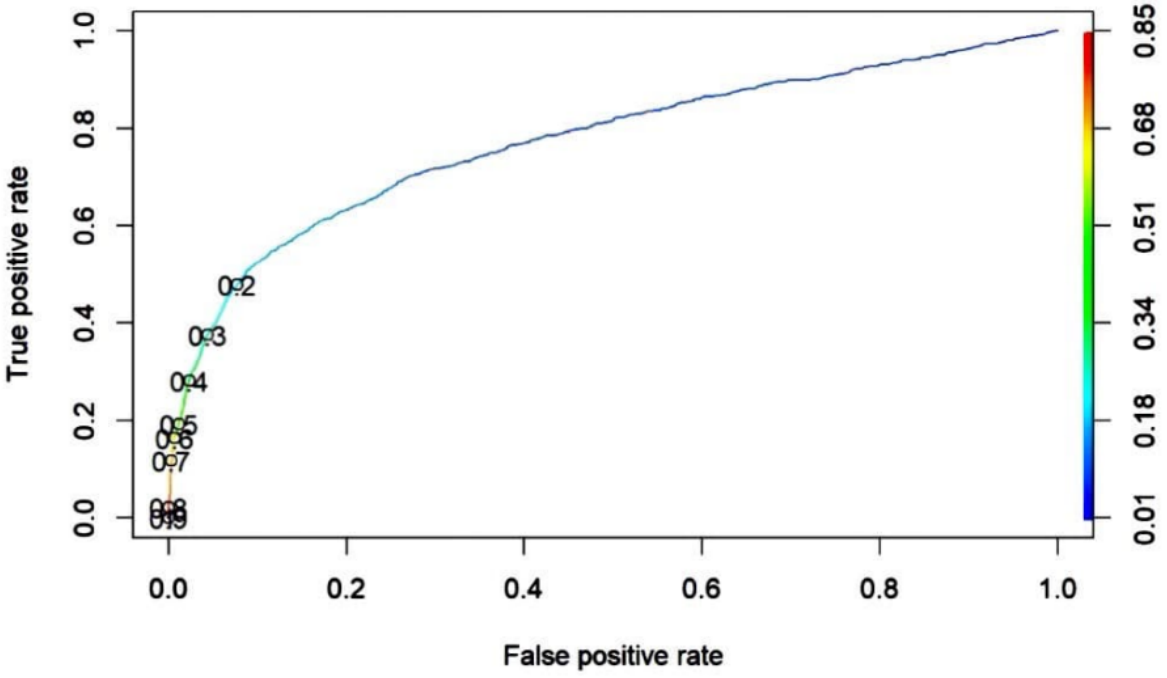
- Conservative Threshold (0.75)
- Balanced Threshold (0.60)

Comparison of Classification Thresholds

Performance Measure	Threshold = 0.75	Threshold = 0.60
Overall Accuracy	89.96%	89.83%
True Negatives	10939	10889
True Positives	111	227
False Positives	25	75
False Negatives	1281	1165

Analysis of Results

- **Accuracy:** The 0.60 cutoff provided a slightly higher overall accuracy of 89.83%.
- **Sensitivity:** Lowering the threshold from 0.75 to 0.60 more than doubled the number of correctly identified subscribers (from 111 to 227), though it increased the number of "wasted" calls (False Positives) from 25 to 75.
- **Operational Impact:** Even at the 0.60 cutoff, the model is highly efficient, filtering out over 10,800 unlikely leads while capturing a significant portion of the potential "yes" responses.



Confusion Matrix

The Confusion Matrix summarizes the prediction results of the final Logistic Regression model.

Actual / Predicted	No	Yes
Actual / No	10889	75
Actual / Yes	1165	227

Interpretation

- ✓ The model correctly classified **10,889** customers who were unlikely to subscribe to the term deposit.
- ✓ It also correctly identified **227** customers who actually subscribed, allowing marketing efforts to focus on high-potential leads.
- ✓ Only **75 customers** were incorrectly predicted as potential subscribers (False Positives), while **1,165 customers** who subscribed were missed by the model (False Negatives).
- ✓ Considering the large imbalance in the dataset, these results indicate that the model performs effectively for real-world telemarketing applications.

Operational Impact

The predictive model offers substantial operational benefits for banking organizations.

Instead of contacting every customer in the database, marketing teams can prioritize customers with the highest predicted probability of subscription.

This approach provides several advantages:

- Reduction in unnecessary telemarketing calls
- Improved utilization of call center resources
- Increased agent productivity
- Lower customer acquisition costs
- Higher campaign conversion rates
- Reduced customer irritation caused by repeated cold calling

By implementing predictive customer scoring, banks can replace traditional mass-calling strategies with intelligent, evidence-based customer selection.

Business Interpretation

- ✓ The results of the Logistic Regression analysis demonstrate that predictive analytics can significantly improve the effectiveness of telemarketing campaigns.
- ✓ The study shows that successful customer conversion depends on the combined influence of demographic characteristics, previous campaign history, communication methods, and macroeconomic conditions rather than any single factor.
- ✓ Customers with favourable economic conditions, successful previous interactions, and appropriate communication channels exhibit a substantially higher likelihood of subscribing to a term deposit.
- ✓ These findings enable marketing managers to allocate resources more efficiently and focus on customers who provide the highest expected return on investment.
- ✓ Rather than relying on intuition or random customer selection, banks can integrate predictive analytics into Customer Relationship Management (CRM) systems to support real-time customer scoring and campaign planning.

Overall Findings

The major findings of the study are summarized below:

- The sequential Logistic Regression approach consistently improved model performance from **Model 1** to **Model 14**, as evidenced by decreasing AIC and Residual Deviance values.
- **Model 14** emerged as the best-performing model with the lowest AIC and Residual Deviance.
- The **Euro Interbank Offered Rate (Euribor)** and **Consumer Price Index (CPI)** were among the most influential macroeconomic predictors.
- Previous campaign success and **pdays** significantly affected customer subscription probability.
- Customers contacted through **cellular communication** demonstrated higher conversion rates than those contacted through traditional telephone channels.
- Demographic variables such as occupation and education also influenced customer decisions.
- A **0.60 probability threshold** provided the best balance between identifying potential subscribers and minimizing unnecessary telemarketing calls.
- The final model achieved an overall classification accuracy of approximately **89.83%**, demonstrating strong predictive performance.

Business Recommendations

Based on the findings of the Binary Logistic Regression model, the following recommendations are proposed to improve the effectiveness of telemarketing campaigns for term deposit subscriptions.

Implement Predictive Customer Scoring

- The study demonstrates that predictive analytics can accurately identify customers with a high probability of subscribing to term deposits.
- Banks should integrate the developed Logistic Regression model into their Customer Relationship Management (CRM) systems to generate customer scores before launching telemarketing campaigns.
- Rather than contacting every customer, marketing teams should prioritize customers with the highest predicted probability of subscription.
- This targeted approach will reduce unnecessary calls, improve agent productivity, and increase campaign conversion rates.

Replace Mass Calling with Intelligent Targeting

Traditional telemarketing campaigns rely heavily on random or mass calling strategies, resulting in significant resource wastage and low conversion rates. The predictive model developed in this study enables banks to identify high-potential customers in advance.

Adopting intelligent customer targeting will:

- ✓ Reduce operational costs
- ✓ Increase successful customer interactions
- ✓ Improve marketing efficiency
- ✓ Enhance overall campaign performance

Utilize Previous Campaign Information

- The analysis identified **previous campaign outcomes** and **pdays (days since the last contact)** as significant predictors of customer subscription.
- Banks should maintain comprehensive customer interaction histories and use these records while planning future telemarketing campaigns.
- Customers with successful previous interactions should receive higher priority during campaign planning.

Optimize Contact Channels

- The findings indicate that customers contacted through **cellular communication** exhibited higher subscription rates than those contacted through traditional telephone channels.
- Banks should therefore prioritize mobile communication wherever feasible and continuously evaluate the effectiveness of different communication channels to maximize customer engagement.

Monitor Economic Indicators

- Macroeconomic variables such as the **Euro Interbank Offered Rate (Euribor)** and **Consumer Price Index (CPI)** significantly influenced customer investment decisions.
- Marketing managers should consider prevailing economic conditions while scheduling telemarketing campaigns.
- During favourable economic periods, promotional activities can be intensified to improve subscription rates.

Continuous Model Improvement

- Customer behaviour changes over time due to market conditions, economic trends, and consumer preferences.
- Therefore, banks should periodically retrain and validate the predictive model using updated customer data.
- Regular model maintenance will ensure sustained predictive performance and improve decision-making accuracy.

Support Data-Driven Marketing Decisions

- Predictive analytics should become an integral part of marketing strategy rather than being used solely as a statistical tool.
- Integrating predictive models with existing CRM and Business Intelligence systems will enable managers to monitor campaign performance, allocate marketing resources effectively, and make evidence-based decisions that improve overall business performance.

Conclusion

- ✓ This study successfully developed a Binary Logistic Regression model to optimize telemarketing campaigns for term deposit subscriptions using predictive analytics.
- ✓ The analysis demonstrated that customer subscription decisions are influenced by a combination of demographic characteristics, previous marketing interactions, communication methods, and macroeconomic conditions.
- ✓ Through a sequential stepwise modeling approach, fourteen logistic regression models were evaluated using **Akaike Information Criterion (AIC)** and **Residual Deviance (RD)**.
- ✓ The final model (Model 14) achieved the best overall performance with the lowest AIC and Residual Deviance values while maintaining statistical reliability.
- ✓ The model achieved an overall prediction accuracy of approximately **89.83%** using a **0.60 probability threshold**, making it highly suitable for practical implementation in banking telemarketing operations.
- ✓ The analysis further revealed that variables such as previous campaign success, contact method, Euribor, Consumer Price Index, and customer demographics significantly influence subscription probability.
- ✓ The findings confirm that predictive analytics provides an effective alternative to traditional mass-calling strategies.
- ✓ By identifying customers with the highest likelihood of subscription, banks can improve marketing efficiency, reduce operational costs, optimize resource allocation, and enhance customer satisfaction.
- ✓ Overall, the study demonstrates that integrating predictive analytics into telemarketing campaigns supports data-driven decision-making and enables financial institutions to achieve higher conversion rates while maintaining efficient marketing operations.

References

1. UCI Machine Learning Repository. (2014). *Bank Marketing Dataset.*
2. Kaggle. (2025). *Bank Marketing Dataset.*
3. Notes - https://archive.ics.uci.edu/ml/datasets/bank+marketing
4. R Core Team. (2025). *R: A Language and Environment for Statistical Computing.* R Foundation for Statistical Computing.
5. RStudio (Posit) official Website https://posit.co/
6. R package documentation (ca Tools, Fast Dummies, ROCR, corrplot, car)
7. Couse notes and study material provided during the IIM Sirmaur EDGE Programme.

APPENDIX

Appendix A – Dataset Preview Include: • First few rows of the Bank Marketing Dataset • Structure of the dataset • Summary statistics	**Appendix B – Data Preprocessing** Include screenshots of: • Data cleaning • Dummy variable creation • Label encoding • Train-test split • Removal of the duration variable
Appendix C – Model Development Include: • R code for Logistic Regression • Model 1 to Model 14 summaries • AIC comparison • Residual Deviance comparison • VIF output	**Appendix D – Model Evaluation** Include: • Confusion Matrix • Accuracy calculation • Threshold comparison (0.75 vs. 0.60) • ROC Curve (if generated) • Probability distribution plots
Appendix E – Visualizations Include charts such as: • Correlation matrix • ROCR curve	